

Spatial Statistics in Epidemiology and Public Health

Lecture 5: Spatial regression

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References

- ▶ Waller and Gotway (2004, Chapter 9) *Applied Spatial Statistics for Public Health Data*. New York: Wiley.
- ▶ Haining, R. (2003). *Spatial Data Analysis: Theory and Practice*. Cambridge: Cambridge University Press.
- ▶ Banerjee, S., Carlin, B.P., and Gelfand, A.E. (2014) *Hierarchical Modeling and Analysis for Spatial Data, 2nd Ed.* Boca Raton, FL: CRC/Chapman & Hall.
- ▶ Blangiardo, M. and Cameletti, M. (2015) *Spatial and Spatio-temporal Bayesian Models with R-INLA*. Chichester: Wiley.

What do we have so far?

- ▶ Tension between *statistical precision* (want large local sample sizes → big regions), and *geographic precision* (want small regions for more detail in map).
- ▶ *Disease mapping* approaches use *small area estimation* techniques to borrow information from all areas and from neighboring areas to improve local estimation in each area.
- ▶ But what about local covariates?
- ▶ Can we adjust for those (say, using regression models)?
- ▶ And still borrow information?
- ▶ With *independent* observations we know how to use *linear* and *generalized linear* models such as linear, Poisson, logistic regression.
- ▶ What happens with *dependent* observations?

Caveat

"...all models are wrong. The practical question is how wrong do they have to be to not be useful."
Box and Draper (1987, p. 74)

What changes with dependence?

- ▶ In statistical modeling, we are often trying to describe the mean of the outcome as a function of covariates, assuming error terms are mutually independent.
- ▶ Where do correlated errors come from?
- ▶ Perhaps outcomes truly correlated (infectious disease).
- ▶ Perhaps we omitted an important variable that has spatial structure itself.

NY leukemia data

- ▶ NY leukemia data and covariates (Waller and Gotway, 2004).
- ▶ 281 census tracts (1980 Census).
- ▶ 8 counties in central New York.
- ▶ 592 cases for 1978-1982.
- ▶ 1,057,673 people at risk.

My first research maps

Some of my first research maps (Waller, 1992)

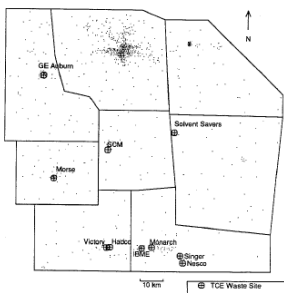


Figure 1.3: Map of New York study area, locations of trichloroethylene hazardous waste sites.

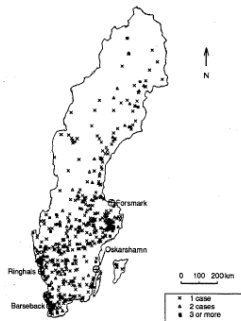
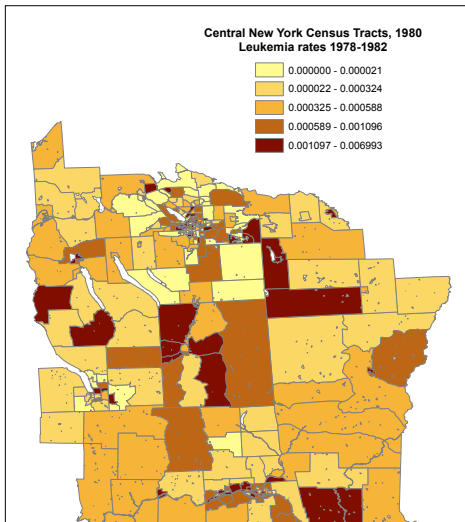
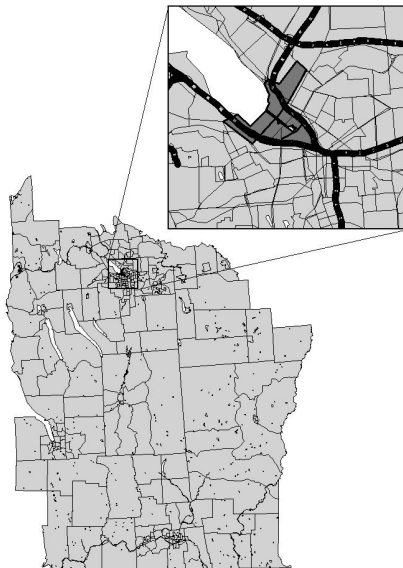


Figure 1.5: Map of Sweden, location of childhood acute lymphocytic leukemia cases 1973-1989.

Crude Rates (per 100,000)



Outliers, where are the top 3 rates?



Building the model: Poisson regression

- ▶ Let Y_i = count for region i .
- ▶ Let E_i = *expected* count for region i .
- ▶ Let $(x_{i,TCE}, x_{i,65}, x_{i,home})$ be the associated covariate values.
- ▶ Poisson regression:

$$Y_i \sim \text{Poisson}(E_i \zeta_i)$$

where

$$\log(\zeta_i) = \beta_0 + x_{i,TCE}\beta_{TCE} + x_{i,65}\beta_{65} + x_{i,home}\beta_{home}.$$

Details

- ▶ Poisson distribution for counts.
- ▶ *Link function*: Natural log of mean of Y_i is a linear function of covariates.
- ▶ β s represent multiplicative increases in expected counts, e^β a measure of relative risk associated with one unit increase in covariate.
- ▶ E_i an *offset*, what we expect if the covariates have no impact.
- ▶ Age, race, sex adjustments in either E_i (standardization) or covariates.

Adding spatial correlation: New York data

- ▶ Assume E_i known, perhaps age-standardized, or based on global (external or internal) rates.
- ▶ Our model is

$$Y_i | \beta, \psi_i \stackrel{ind}{\sim} \text{Poisson}(E_i \exp(\mathbf{x}'_i \beta + \psi_i)),$$

$$\log(\zeta_i) = \beta_0 + x_{i,TCE} \beta_{TCE} + x_{i,65} \beta_{65} + x_{i,home} \beta_{home} + \psi_i.$$

- ▶ The ψ_i represent the *random intercepts*.
- ▶ Add *overdispersion* via $\psi_i \stackrel{ind}{\sim} N(0, \nu_\psi)$.
- ▶ Add spatial correlation via

$$\psi \sim MVN(\mathbf{0}, \Sigma).$$

Priors and “shrinkage”

- ▶ Overdispersion model (i.i.d. ψ_i) results in each estimate being a compromise between the *local* SMR and the *global average* SMR.
- ▶ “Borrows information (strength)” from other observations to improve precision of local estimate.
- ▶ “Shrinks” estimate toward global mean. (Note: “shrink” does not mean “reduce”, rather means “moves toward”).

Local shrinkage

- ▶ Spatial model (correlated ψ_i) results in each estimate begin a compromise between the *local* SMR and the *local average* SMR.
- ▶ Shrinks each ψ_i toward the average of its *neighbors*.
- ▶ Can also include *both* global and local shrinkage (Besag, York, and Mollié 1991).
- ▶ How do we fit these models?

Bayesian inference

Bayesian inference regarding model parameters based on *posterior distribution*

$$Pr[\beta, \psi | \mathbf{Y}]$$

proportional to the product of the likelihood times the prior

$$Pr[\mathbf{Y} | \beta, \psi] Pr[\psi] Pr[\beta].$$

Defers spatial correlation to the prior rather than the likelihood.

Spatial priors

- ▶ Could model *joint* distribution

$$\psi \sim MVN(\mathbf{0}, \Sigma).$$

- ▶ Could also model *conditional* distribution

$$\psi_i | \psi_{j \neq i} \sim N \left(\frac{\sum_{j \neq i} c_{ij} \psi_j}{\sum_{j \neq i} c_{ij}}, \frac{1}{v_{CAR} \sum_{j \neq i} c_{ij}} \right), i = 1, \dots, N.$$

where c_{ij} are *weights* defining the neighbors of region i .

- ▶ Adjacency weights: $c_{ij} = 1$ if j is a neighbor of i .

CAR priors

- ▶ The conditional specification defines the *conditional autoregressive* (CAR) prior (Besag 1974, Besag et al. 1991).
- ▶ Under certain conditions on the c_{ij} , the CAR prior defines a valid multivariate joint Gaussian distribution.
- ▶ Variance covariance matrix a function of the *inverse* of the matrix of neighbor weights.

Fitting Bayesian models

- ▶ Posterior often difficult to calculate mathematically.
- ▶ Markov chain Monte Carlo: Iterative simulation approach to model fitting.
- ▶ Given *full conditional* distributions, simulate a new value for each parameter, holding the other parameter values fixed.
- ▶ The set of simulated values converges to a sample from the posterior distribution.
- ▶ Alternative: *integrated nested Laplace analysis* using the `inla` package (example code).

Complete model specification

$$Y_i | \beta, \psi_i \stackrel{ind}{\sim} \text{Poisson}(E_i \exp(\mathbf{x}'_i \beta + \psi_i)),$$

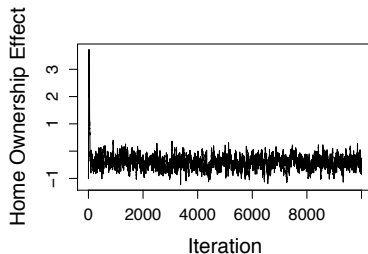
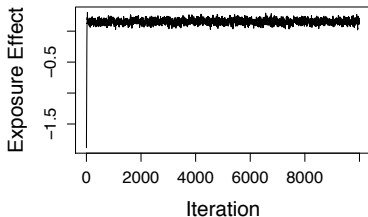
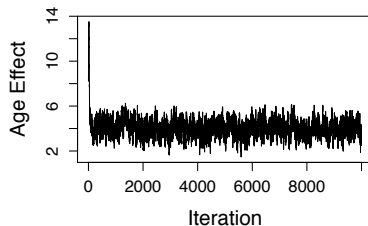
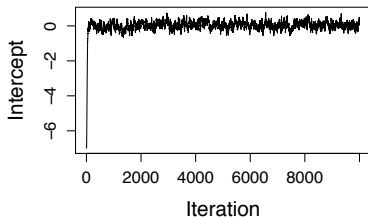
$$\log(\zeta_i) = \beta_0 + x_{i,TCE} \beta_{TCE} + x_{i,65} \beta_{65} + x_{i,home} \beta_{home} + \psi_i.$$

$$\beta_k \sim \text{Uniform}.$$

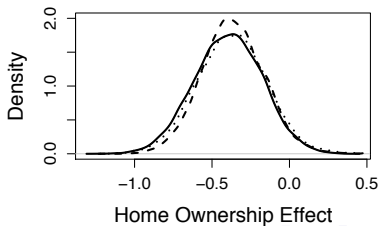
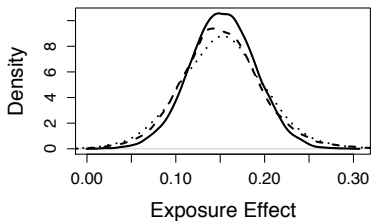
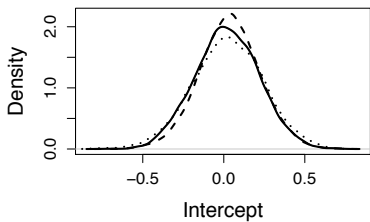
$$\psi_i | \psi_{j \neq i} \sim N \left(\frac{\sum_{j \neq i} c_{ij} \psi_j}{\sum_{j \neq i} c_{ij}}, \frac{1}{v_{CAR} \sum_{j \neq i} c_{ij}} \right), i = 1, \dots, N.$$

$$\frac{1}{v_{CAR}} \sim \text{Gamma}(0.5, 0.0005).$$

MCMC trace plots



Posterior densities



MCMC posterior estimates

Covariate	Posterior Median	95% Credible Set
β_0	0.048	(-0.355, 0.408)
β_{65}	3.984	(2.736, 5.330)
β_{TCE}	0.152	(0.066, 0.226)
β_{home}	-0.367	(-0.758, 0.049)

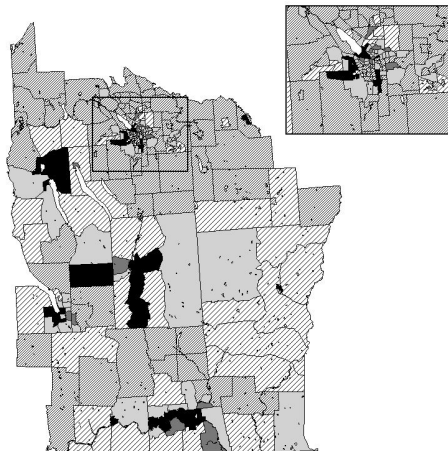
But there's more!

- ▶ A nifty thing about MCMC estimates:

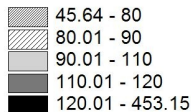
We get posterior samples from any function of model parameters by taking that function of the sampled posterior parameter values.

- ▶ Gives us posterior inference for $SMR_i = Y_{i,fit}/E_i$.
- ▶ Also can get $Pr[SMR_i > 200 | \mathbf{Y}]$ and map these *exceedence probabilities*.

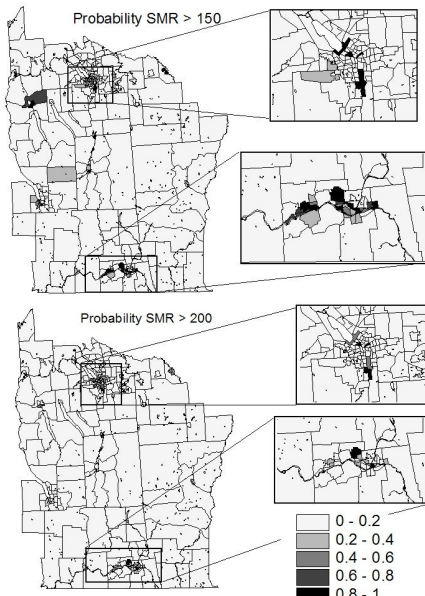
Posterior median SMRs



Posterior median local SMR
CAR prior



Posterior exceedence probabilities



What do we have?

- ▶ Associations between local covariates and local outcomes (counts and rates).
- ▶ Spatial correlation between random intercepts (inside the link function).
- ▶ (Aside: This is a clever idea since we can use a multivariate Gaussian distribution for correlation...).
- ▶ Result: Local rates adjusted for covariates *and* smoothed by borrowing information.
- ▶ Many examples in the literature, and many extensions!

Conclusions

- ▶ What method to use depends on what data you have and what question you want to answer.
- ▶ All methods try to balance trend (fixed effects) with correlation (here, with random effects).
- ▶ All models wrong, some models useful.
- ▶ Trying more than one approach often sensible.

Slippery Slopes: What if *slopes* vary in space?

- ▶ Collaborators: Paul Gruenewald, Dennis Gorman, Li Zhu, Carol Gotway, and David Wheeler
- ▶ References:
 - ▶ Waller et al. (2008) Quantifying geographical associations between alcohol distribution and violence... *Stoch Environ Res Risk Assess* **21**: 573-588.
 - ▶ Wheeler and Caldor (2009) As assessment of coefficient accuracy... *J Geogr Systems* **9**: 573-588.
 - ▶ Wheeler and Waller (2009) Comparing spatially varying coefficient models... *J Geogr Systems* **11**: 1-22.
 - ▶ Finley (2011) Comparing spatially-varying coefficient models... *Methods in Ecology and Evolution* **2**: 143-154.

What do we want to do?

- ▶ Quantify associations between outcomes and covariates as observed in data.
- ▶ Adjust for spatial correlation (spatial regression) using a random intercept with a CAR prior.
- ▶ What if strength of association varies across space?
- ▶ Usually, we assume β is the same at every location, what if it varies (but is spatially correlated)?
- ▶ Can we have a *random slope*? Can we use CAR priors for that?

What about spatially varying associations?

- ▶ Fix it: Geographically weighted regression (GWR)
 - ▶ Fotheringham et al. (2002)
- ▶ Model it: Spatially varying coefficient (SVC) models
 - ▶ Leyland et al. (2000), Assuncao et al. (2003), Gelfand et al. (2003), Gamerman et al. (2003), Congdon (2003, 2006)

Data example: Alcohol, illegal drugs, violent crime

- ▶ Outcome: Rates (number of cases per person per year) of violent crimes (police/sheriff reports).
- ▶ Covariates: Alcohol distribution (licenses and sales), illegal drug arrests (police/sheriff reports).
- ▶ Potential confounders: Sociodemographics (census).
- ▶ Linked to common spatial framework (census tracts) via GIS.

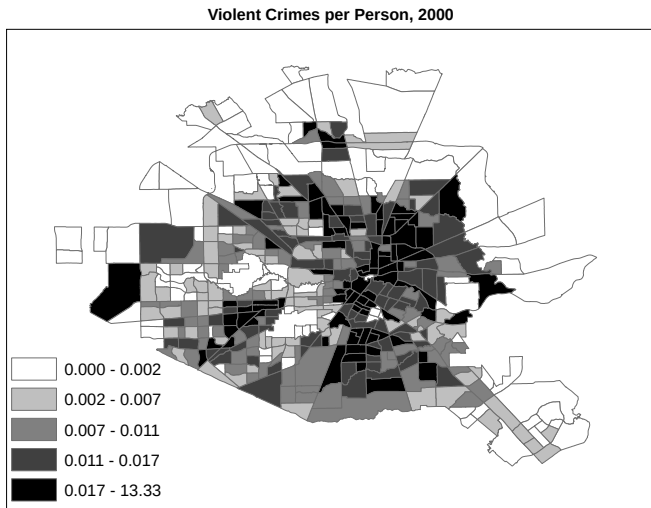
Translation complications

- ▶ When are crime data like disease data?
 - ▶ Counts from small areas.
 - ▶ Per person “rate” of interest.
- ▶ When are crime data not like disease data?
 - ▶ Outcome not as “rare”.
 - ▶ Police vs. medical records.
 - ▶ Residents not only ones at risk.

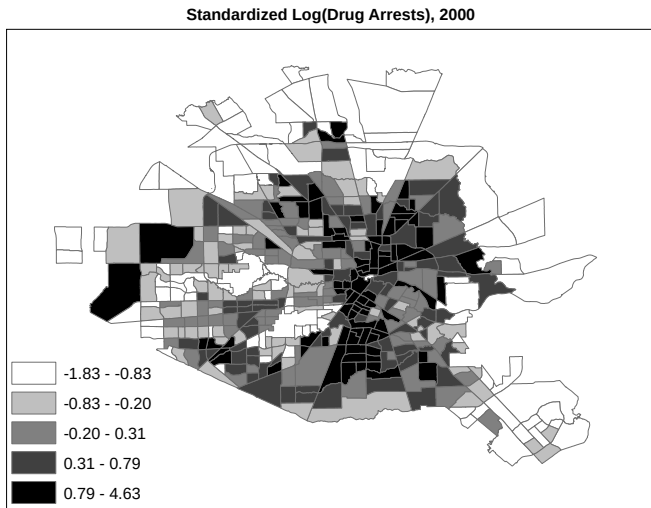
Data description

- ▶ Spatial support: 439 census tracts (2000 Census).
- ▶ Violent crime (murder, robbery, rape, aggravated assault) “first reports” for year 2000 from City of Houston Police Department website.
- ▶ Gorman et al. (2005, *Drug Alcohol Rev*) report less than 5% discrepancy with 2000 Uniform Crime Reports.
- ▶ 98% of reports geocoded to the census tract level.
- ▶ Alcohol data (locations of active distribution sites in 2000) from Texas Alcoholic Beverage Commission (6,609 outlets), 99.5% geocoded to the tract level.
- ▶ Drug law violations (also from City of Houston police data). 98% geocoded to the tract level.

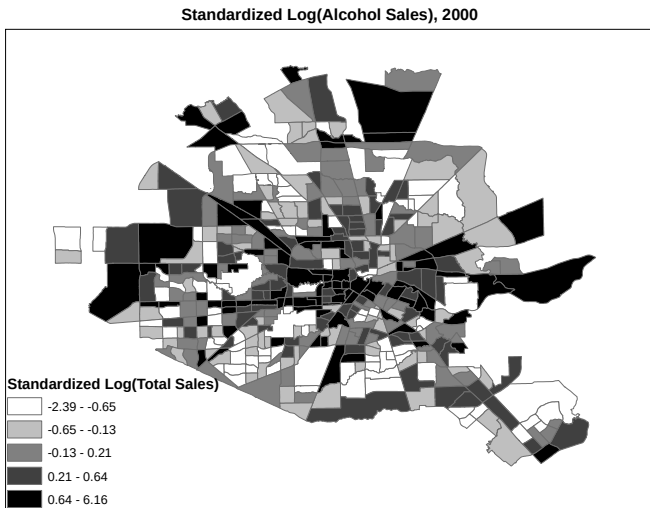
Violent Crime reporting rates, Houston, 2000



Standardized log(drug arrests), Houston, 2000



Standardized $\log(\text{alcohol sales})$, Houston, 2000

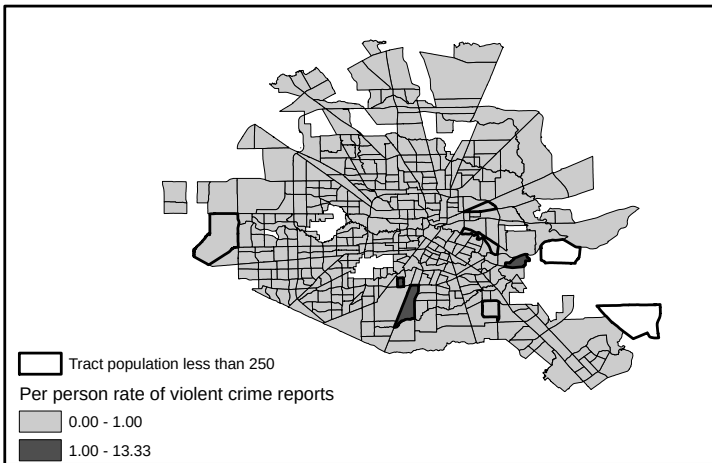


Data “features”

- ▶ 7 of 439 tracts have extremely small population sizes: 1, 3, 4, 16, 34, 116, and 246.
- ▶ Tracts typically have 3,000-5,000 residents.
- ▶ Local rates for such tracts are extremely unstable (e.g., 40 reports, 3 residents).
- ▶ Actually a motivating a reason for including the spatially varying intercept: borrow information across regions.

Low population tracts and high rates

Low population size tracts



Basic Poisson regression

- ▶ Let Y_i = number of reports in tract i , $i = 1, \dots, 439$.
- ▶ Suppose $Y_i \sim \text{Poisson}(E_i \exp(\mu_i))$, where E_i = the “expected” number of reports under some null model.
- ▶ Typically, $E_i = n_i R$ where all n_i individuals in region i are equally likely to report.
- ▶ $\exp(\mu_i)$ = “relative risk” of outcome in region i .
- ▶ We add covariates in linear format (within $\exp(\cdot)$):
$$\mu_i = \beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i}.$$
- ▶ Same “skeleton” for both GWR and SVC.

Why do we have E_i ?

- $E_i = n_i R$ represents an “offset” in the model and lets us use Poisson regression to model *rates* as well as *counts*.

$$\begin{aligned} E[Y_i] &= E_i \exp(\beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i}) \\ &= \exp(\ln(E_i) + \beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i}) \\ &= \exp(\ln(n_i) + \ln(R) + \beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i}) \\ \log(E[Y_i]) &= \ln(n_i) + \ln(R) + \beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i} \end{aligned}$$

- GWR offset: $\ln(n_i)$, SVC offset: $\ln(n_i) + \ln(R)$.

GWPR (Nakaya et al., 2005)

- ▶ Geographically weighted Poisson regression.
- ▶ $\hat{\beta}_{GWPR} = (\mathbf{X}'\mathbf{W}(\mathbf{s})\mathbf{A}(\mathbf{s})\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}(\mathbf{s})\mathbf{A}(\mathbf{s})\mathbf{Z}(\mathbf{s})$.
- ▶ $\mathbf{A}(\mathbf{s})$ = diagonal matrix of Fisher scores.
- ▶ $\mathbf{Z}(\mathbf{s})$ = Taylor-series approximation to transformed outcomes.
- ▶ Update $\mathbf{A}(\mathbf{s})$, $\mathbf{Z}(\mathbf{s})$ and $\hat{\beta}_{GWPR}$ until convergence.

Fitting in *R*

- ▶ Waller et al. (2007) use GWR 3.0 software.
- ▶ In *R*: `maptools` will read in ArcGIS-formatted shapefile (files) into *R*.
- ▶ `spgwr` fits linear GWR and GLM-type GWR.

SVC

- ▶ $\mu_i = \beta_0 + \beta_1 x_{alc,i} + \beta_2 x_{drug,i} + b_{1,i} x_{alc,i} + b_{2,i} x_{drug,i} + \phi_i + \theta_i$.
- ▶ $\beta_0, \beta_1, \beta_2 \sim \text{Uniform}$.
- ▶ Random intercept has 2 components (Besag et al. 1991):

$$\theta_i \stackrel{\text{ind}}{\sim} N(0, \tau^2)$$

$$\phi_i | \phi_j \sim N \left(\frac{\sum_j w_{ij} \phi_j}{\sum_j w_{ij}}, \frac{1}{\lambda \sum_j w_{ij}} \right).$$

where w_{ij} defines neighbors, and λ controls spatial similarity.

- ▶ θ_i allows overdispersion (smoothing to global mean).
- ▶ ϕ_i follows conditionally autoregressive distribution (smoothing to local mean), generates MVN but more convenient for MCMC.

Defining the SVCs

- ▶ $\mathbf{b}_1, \mathbf{b}_2$ also given spatial priors and allowed to be correlated with one another.
- ▶ We use a formulation by Leyland et al. (2000) which defines

$$(b_{1,i}, b_{2,i})' \sim MVN((0, 0)', \Sigma)$$

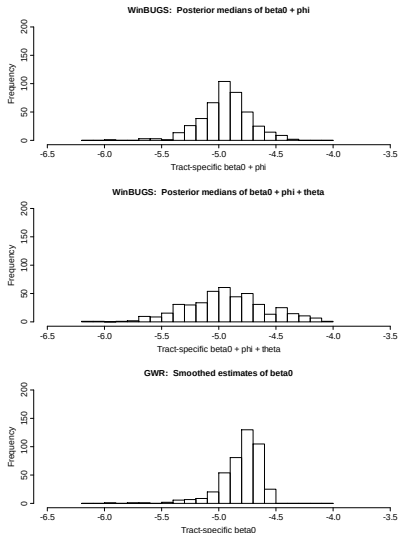
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- ▶ Typically fit via MCMC (but often runs slowly).

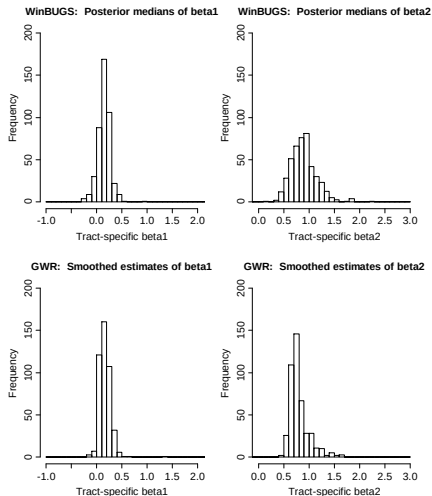
Implementation

- ▶ Example code using `spgwr` library.)
 - ▶ Convergs in minutes.
- ▶ MCMC to fit SVC model.
 - ▶ Converged in hours.
- ▶ Fit several versions of SVC model and compared fit via deviance information criterion (Spiegelhalter et al., 2003).
- ▶ Best fit included spatial varying coefficients, random intercept, and correlation between alcohol and drug effects.

Results: Intercept

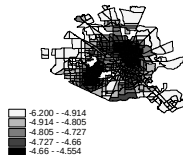


Results: Alcohol sales and drug arrests

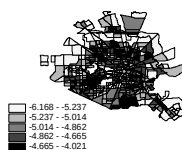


Estimated effects

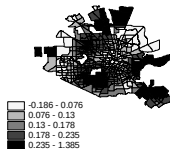
GWR: Local Estimate of Intercept



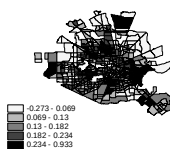
WinBUGS: Local Estimate of Intercept



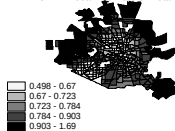
GWR: Local Estimate of Beta 1



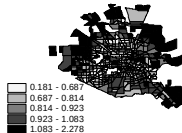
WinBUGS: Local Estimate of Beta 1



GWR: Local Estimate of Beta 2



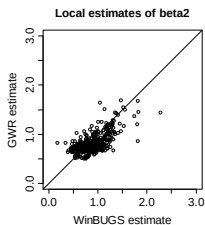
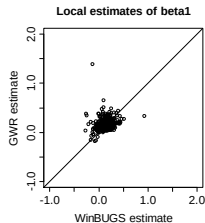
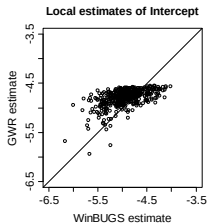
WinBUGS: Local Estimate of Beta 2



Similarities

- ▶ Alcohol: Increased impact in western, south-central, and southeastern parts of city.
- ▶ Illegal drug: Increased impact on periphery, lower influence in central and southwestern parts of city.
- ▶ Intercept: Increased risk of violence in central area, above and beyond that predicted by alcohol sales and illegal drug arrests.
- ▶ But, associations not too close...

Results: tract-by-tract



Differences

- ▶ GWR much smoother based on global best fit for bw .
- ▶ SVC used adjacency-based smoothing and a different amount of smoothing for each covariate.
- ▶ GWR: collinearity between surfaces (Wheeler and Tiefelsdorf, 2005).
- ▶ SVC: Model based approach removes (or at least reduces) collinearity.

We will try it out this afternoon.

- ▶ Houston data on violent crime, alcohol sales, and illegal drug arrests.
- ▶ ArcGIS shapefile.
- ▶ Required R libraries: `maptools` (to read in shape file), `RColorBrewer` (to set colors), `classInt` (to set intervals of values for mapping), and `spgwr` (for GWR).

Conclusions

- ▶ GWR and SVC very different approaches to the same problem.
- ▶ Qualitatively similar in results, but not directly transformable.
- ▶ GWR fixed problems within somewhat of a black box.
- ▶ SVC allows probability model-based inference with lots of flexibility but at a computational cost (both in set-up and implementation).
- ▶ Further research:
 - ▶ Wheeler and Waller (2009): Attempt to set up SVC model to more closely mirror amount of smoothing in GWR.
 - ▶ Collinearity “ribbons”.
 - ▶ Griffith (2002) eigenvector spatial filtering to adjust collinearity. Interpretability?